6 A capacitor of capacitance 47 μF, a cell of electromotive force (e.m.f.) 1.50 V and a resistor R are connected into the circuit of Figure 6.1.

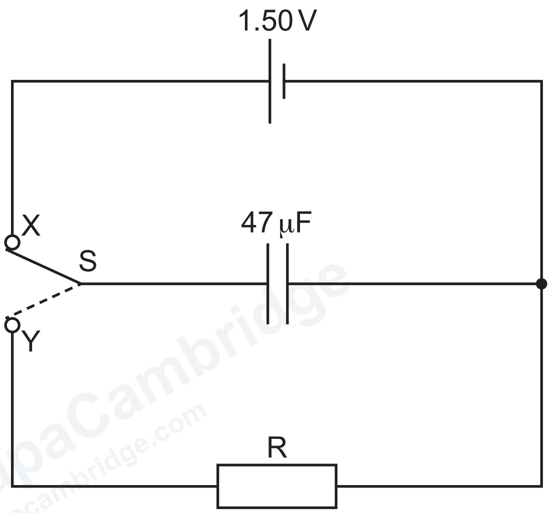

Figure 6.1

The two-way switch S alternates between terminals X and Y with a frequency of 0.20 Hz, as shown in the variation of switch position with time *t* in Figure 6.2.

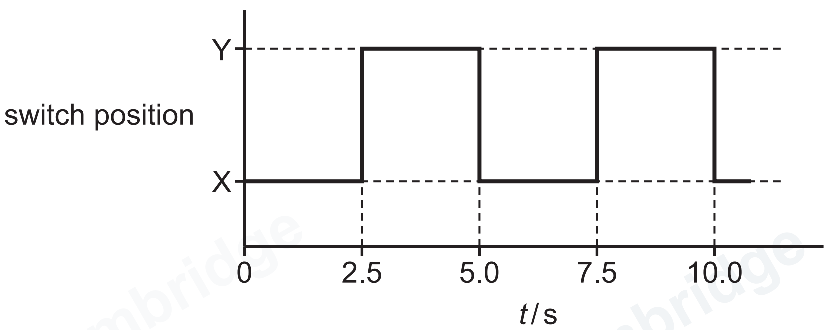

Figure 6.2

The time constant of the discharge circuit is 2.4 s.

(a) **(i)** Calculate the resistance, in kΩ, of R.

resistance = ... kΩ [2]

(ii) Show that the minimum potential difference (p.d.) across the capacitor is 0.53 V.

[2]

(b) On Figure 6.3, sketch the variation of the p.d. V_C across the capacitor with t between $t = 0$ and $t = 10.0$ s.

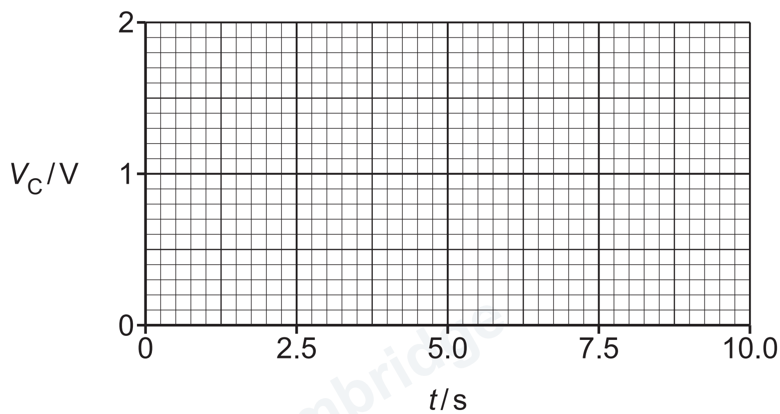

Figure 6.3

[3]

(c) The frequency at which the switch alternates is now doubled.

Explain, without calculation, whether the minimum p.d. across the capacitor is now less than, equal to or greater than 0.53 V.

....

....

.... [2]

[Total: 9]